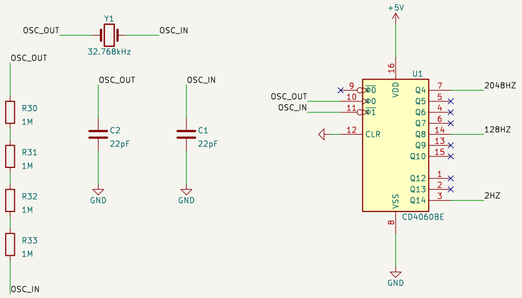
Logic-chip clock (HH:MM) — no microcontroller

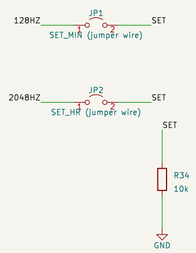
Net labels with the same name are connected. Reference designators match bom.csv.

1. Timebase: 32.768 kHz / 16384 = 2 Hz



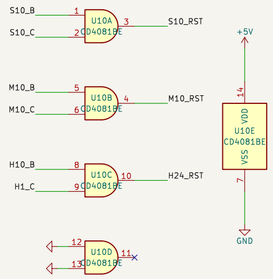
R30–R33: 4 x 1M in series = 4M feedback resistor

Time-setting (jumper wire)



No buttons: touch a spare jumper wire across JP1 (~1 min/s) or JP2 (~1 hr per 3.5 s)

3. Reset logic (CD4081)

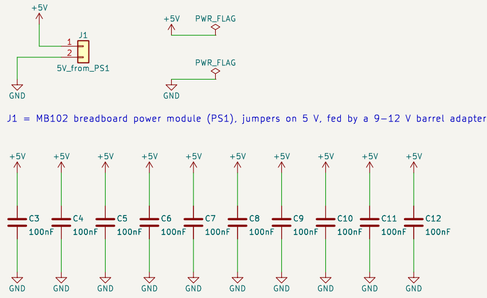


U10A: seconds-tens = 6 -> reset to 0

U10B: minutes-tens = 6 -> reset to 0

U10C: hours = 24 -> reset both to 00

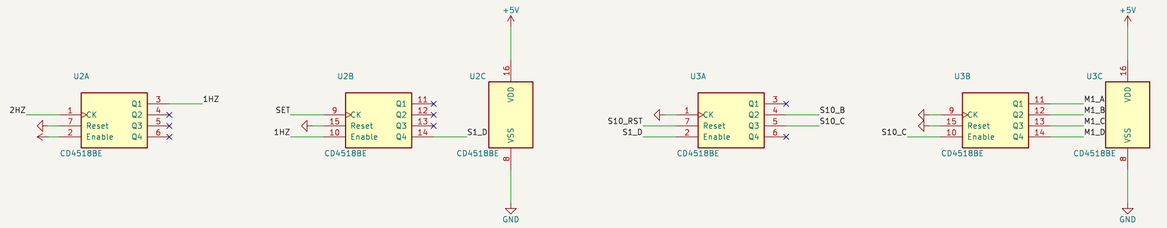
5. Power + decoupling



J1 = MB102 breadboard power module (PS1), jumpers on 5 V, fed by a 9–12 V barrel adapter

C3–C12: one 100nF beside each chip (U1–U10)

2. Counter chain (CD4518 dual BCD counters)



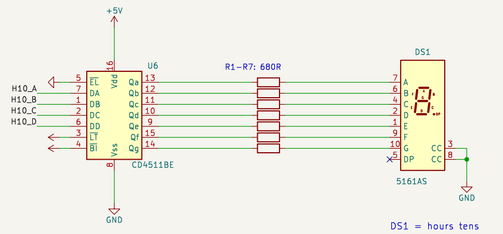
U2: A: divide by 2 (2 Hz -> 1 Hz) B: seconds ones

U3: A: seconds tens B: minutes ones

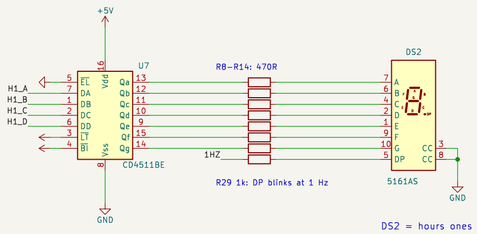
U4: A: minutes tens B: hours ones

U5: A: hours tens B: unused

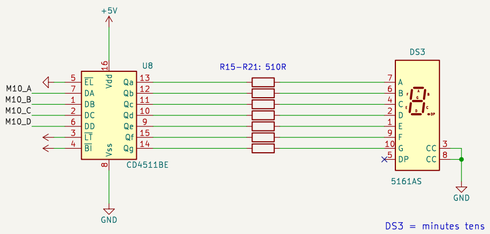
4. Decoders + displays (CD4511 -> 470–680R per segment -> 5161AS common cathode)



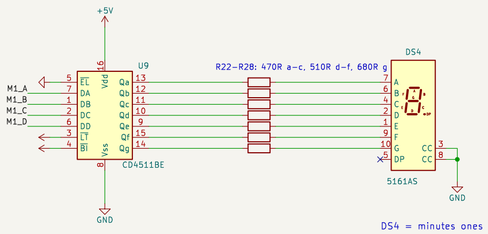
DS1 = hours tens



DS2 = hours ones



DS3 = minutes tens



DS4 = minutes ones

Notes

- A digit ticks when the previous digit rolls over: ones use Q4 falling (9->0), tens use Q3 falling (6->0 reset).
- Seconds are counted (U2B, U3A) but not displayed. After power-up, bridge JP2 (hours) then JP1 (minutes) with a jumper wire.
- Every unused CMOS input is tied to +5V or GND. Unused outputs are marked no-connect.
- Small sideways arrow = +5V, small sideways ground = GND (values hidden to keep pin rows readable).
- DS1–DS4 use the KiCad SM420561N symbol: same 10-pin common-cathode pinout as the 5161AS. Check your part.
- Breadboard build: footprints are filled in only so this can become a PCB later.

Budget build: no buttons, mixed 470/510/680R segments, 4x1M feedback. Refs match bom.csv
CD4060 timebase, CD4518 counters, CD4081 resets, CD4511 decoders

Sheet: /	
File: logic-clock.kicad_sch	
Title: Logic-chip clock (HH:MM)	
Size: A2	Date: 2026-10-31
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: B**
	Id: 1/1